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Information Retrieval: Movies and Series

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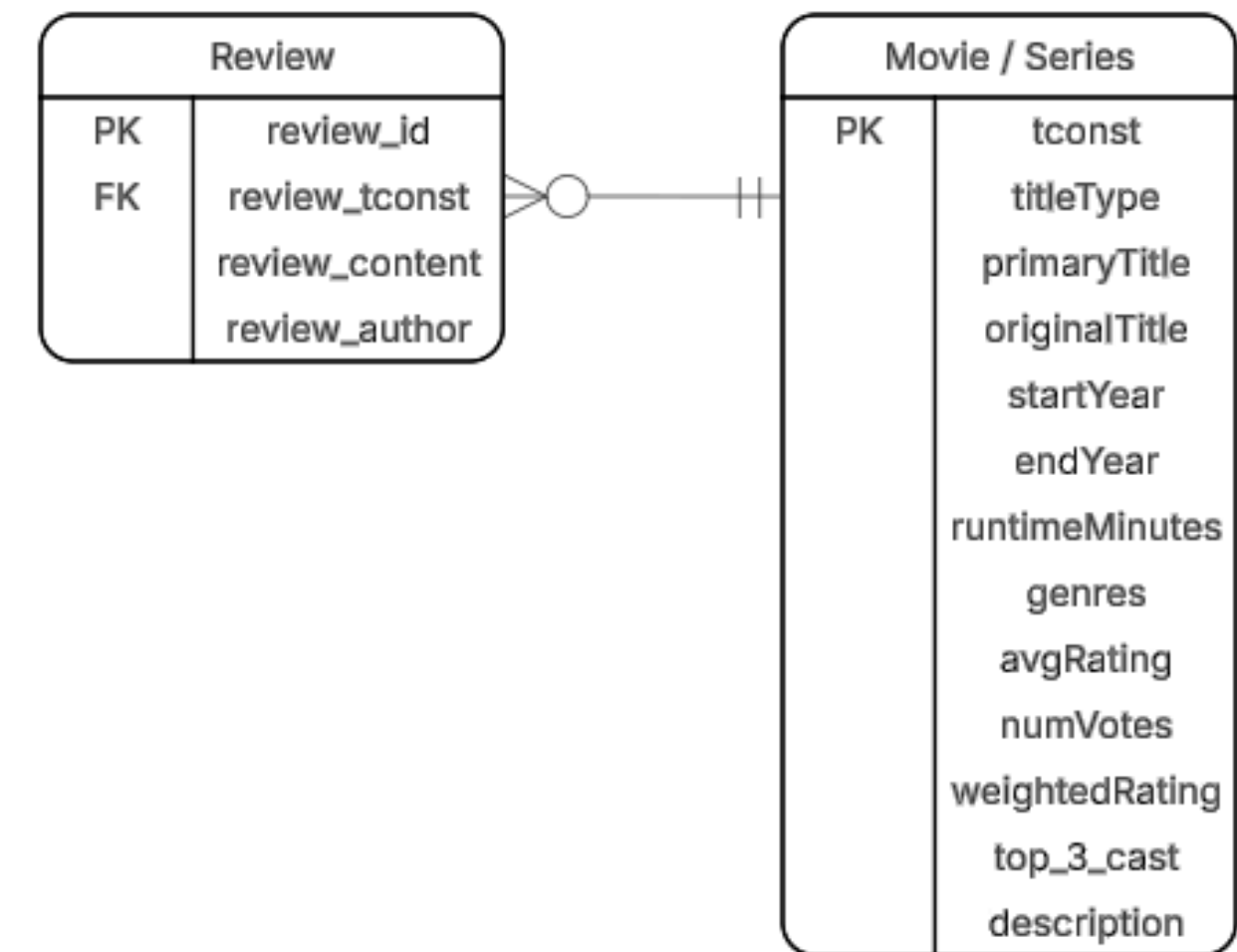
Overview

Sources

Data extracted from IMDb, Kaggle, OpenDataBay and StreamWithVPN

Content

Movies and Series → Documents with structured and unstructured data



Data Characterization

- Each document encompasses **multiple attributes**: primary and original titles, content type, release/end years, duration, genre, rating metrics (mean rating, vote count, weighted rating), principal cast members, and description.
- Our **main text source** is the description of each movie/serie document.

```
{
  "tconst": "tt0468569",
  "titleType": "movie",
  "primaryTitle": "The Dark Knight",
  "originalTitle": "The Dark Knight",
  "startYear": 2008,
  "endYear": null,
  "runtimeMinutes": 152,
  "genres": [
    "Action",
    "Crime",
    "Drama"
  ],
  "averageRating": 9.1,
  "numVotes": 3079402,
  "weightedRating": 9.091059698803372,
  "top_3_cast": [
    "Christian Bale",
    "Christian Bale - Bruce Wayne",
    "Heath Ledger",
    "Heath Ledger - Joker",
    "Aaron Eckhart",
    "Aaron Eckhart - Harvey Dent"
  ],
  "description": "Batman raises the stakes in his war on crime. With the help o
},
```

Solr Setup and Configuration

- Apart from the default schema automatically created by Solr when using the *data_driven_schema_config*, we created a different schema, for better customization.
- Our **custom schema** uses different index and query analyzers, specifying which is included in each field.

| Indexing and Filters

Basic Schema

Uses general settings and types to fields.

Custom Schema

- **title_text** uses StandardTokenizerFactory, LowerCaseFilterFactory, ASCIIFoldingFilterFactory, ...
- **name_text** uses StandardTokenizerFactory, ShingleFilterFactory.
- Query analysis incorporates SynonymGraphFilterFactory for plot descriptions and title fields

Field	Type	Indexed
primaryTitle	title_text	yes
originalTitle	title_text	yes
titleType	string_ci	yes
startYear	pint	yes
endYear	pint	yes
runtimeMinutes	pint	yes
genres	string_ci	yes
averageRating	pfloat	yes
numVotes	pint	yes
weightedRating	pfloat	yes
top_3_cast	name_text	yes
top_3_cast_search	text_general	yes
description	text_general	yes

Information Need: *"Funny TV Show"*

Context: The objective with this query was to find series included in the comedy and humor genre, without specifying those keywords directly. We also added a time frame and ordered by score.

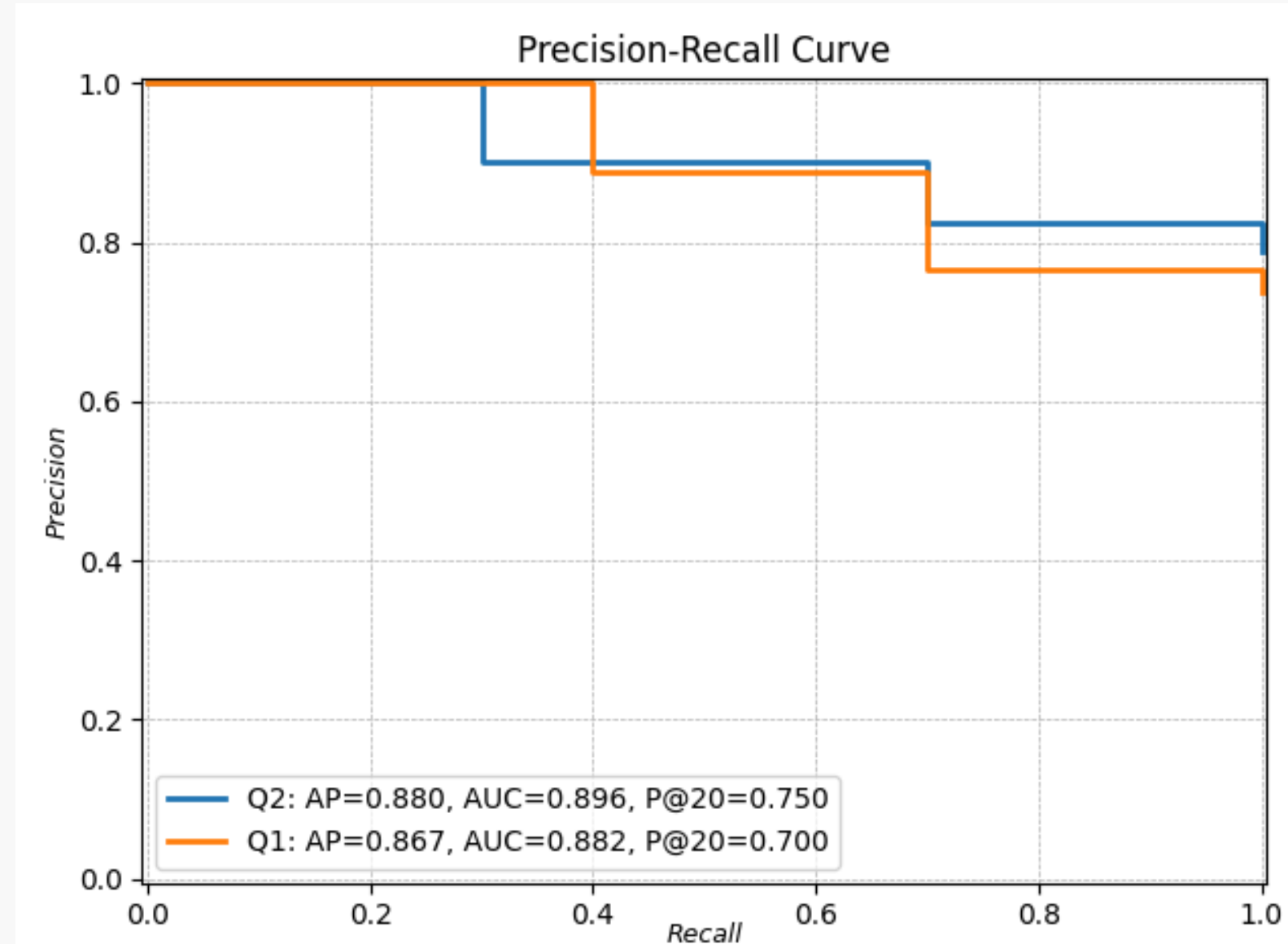
Baseline	Boosted
"query": "Funny TV Show"	"query": "Funny TV Show"
"qf": ["primaryTitle", "description", "genres", "top_3_cast"]	"qf": "primaryTitle^2 genres^2 description^3 top_3_cast^2"
"fq": ["titleType:tvSeries", "startYear:[2000 TO 2015]"]	"fq": ["titleType:tvSeries", "startYear:[2000 TO 2015]"]
	"pf": "description^6"
	"ps": 3

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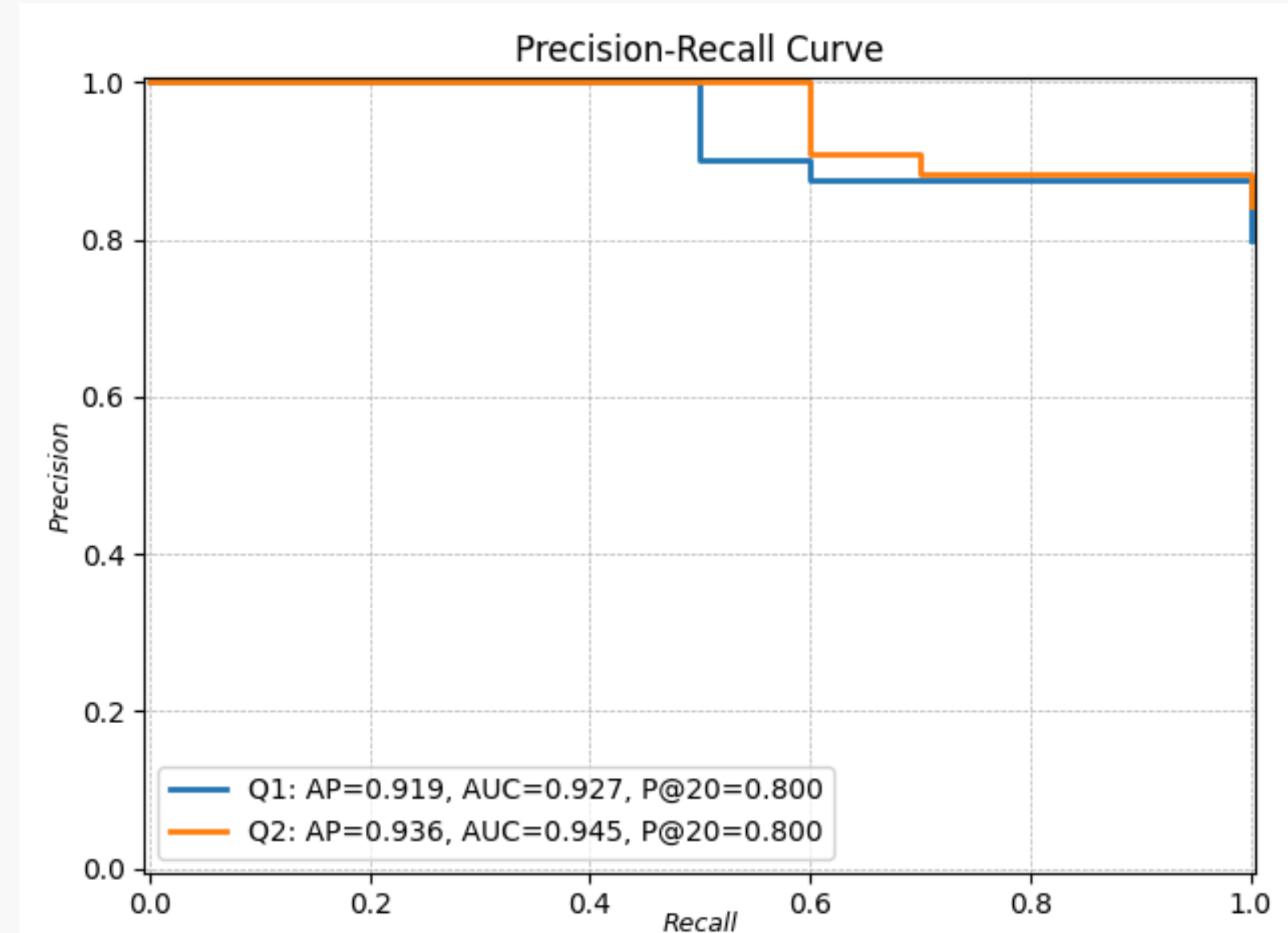
Queries and Retrieval

Information Need: "Funny TV Show"

Basic Schema



Custom Schema



Information Need: *“Robert De Niro or Al Pacino Crime Drama Themed Movies”*

Context: The objective with this query was to search for movies containing either one of both the actors specified, but with both genres present. We again added a rating delimitation and sorted by score.

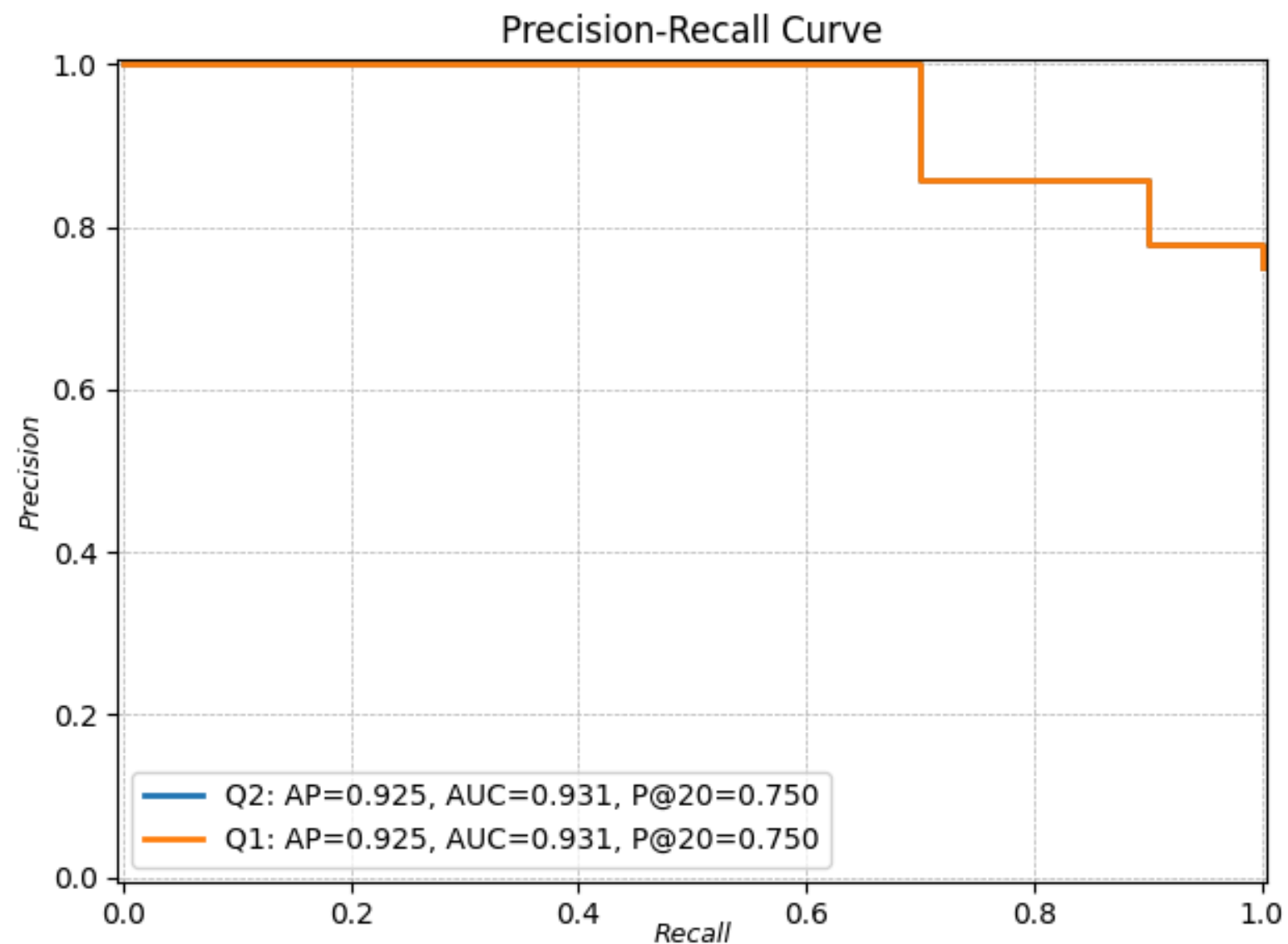
Baseline	Boosted
"query": "\"Robert De Niro\" or \"Al Pacino\" Crime Drama"	"query": "\"Robert De Niro\" or \"Al Pacino\" Crime Drama"
"qf": ["primaryTitle", "description", "genres", "top_3_cast"]	"qf": "primaryTitle^2 genres^2 description^3 top_3_cast^2"
"fq": ["titleType:movie", "weightedRating:[7.5 TO 10]"]	"fq": ["titleType:movie", "weightedRating:[7.5 TO 10]"]
	"pf": "description^6"
	"ps": 3

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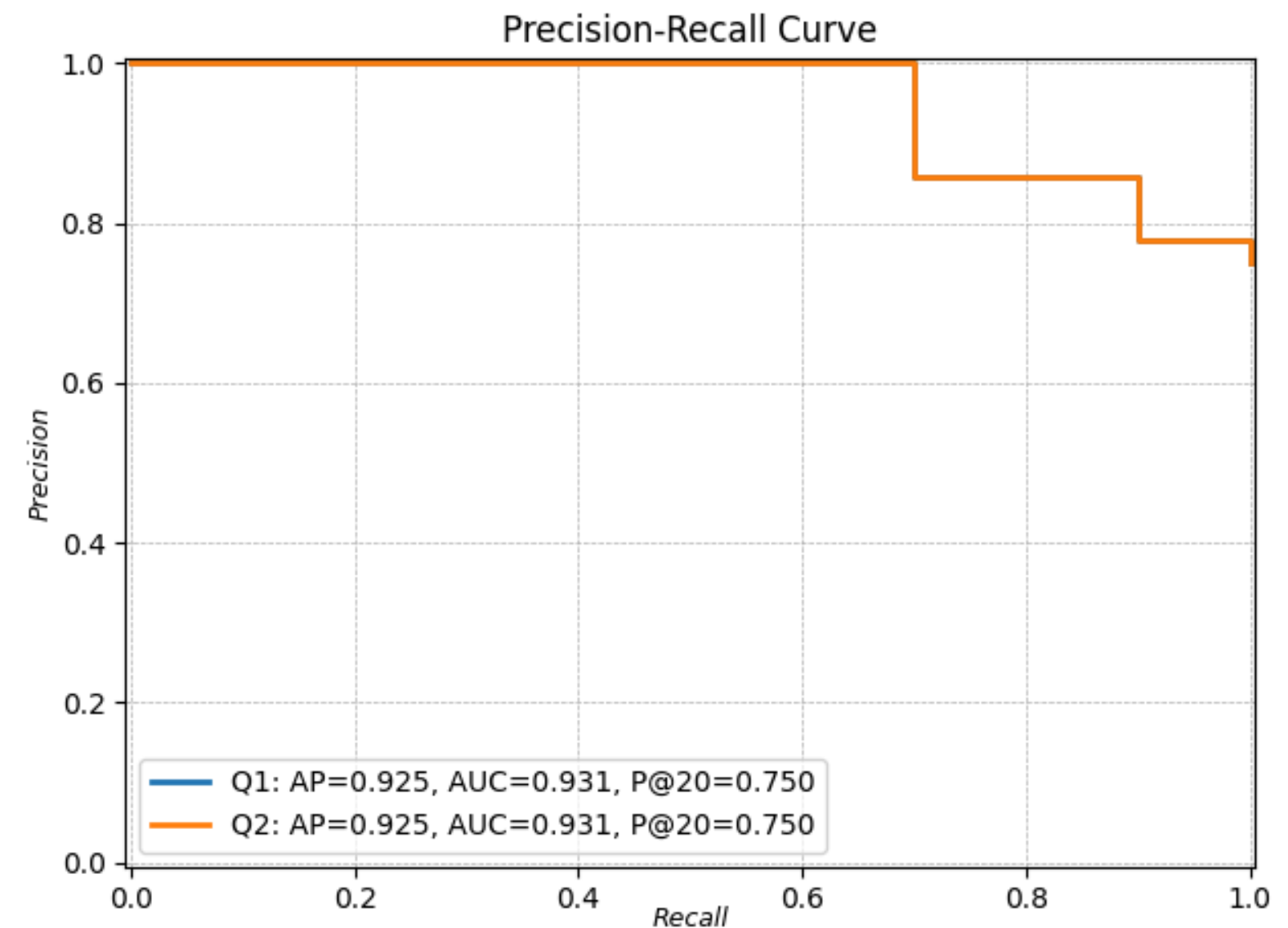
Queries and Retrieval

Information Need: *"Robert De Niro or Al Pacino Crime Drama Themed Movies"*

Basic Schema



Custom Schema



Information Need: *“Artificial Intelegence / Sentient Robots Themed Movies”*

Context: The objective with this final query was to know which movies were related to the themes specified, without adding specific details. The objective was to further test how the retrieval system worked when presented with general themes.

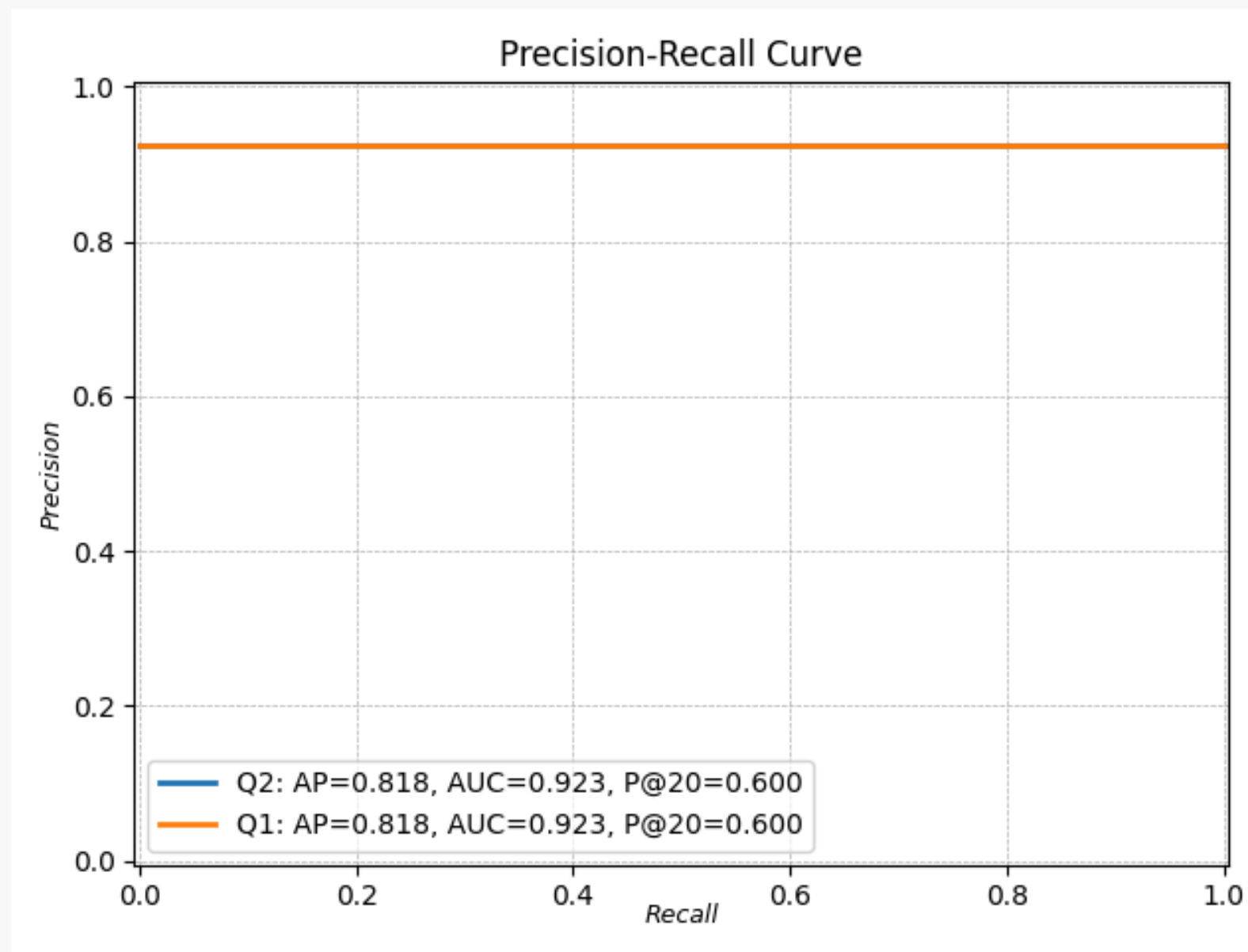
Baseline	Boosted
"query": "\"artificial intelligence\" AI robots androids \"machine learning\" \"sentient machines\""	"query": "\"artificial intelligence\" AI robots androids \"machine learning\" \"sentient machines\"",
"qf": ["primaryTitle", "description", "genres", "top_3_cast"]	"qf": "primaryTitle^2 genres^2 description^3 top_3_cast^2"
"fq": ["titleType:movie"]	"fq": ["titleType:movie"]
	"pf": "description^6"
	"ps": 3

/07.3

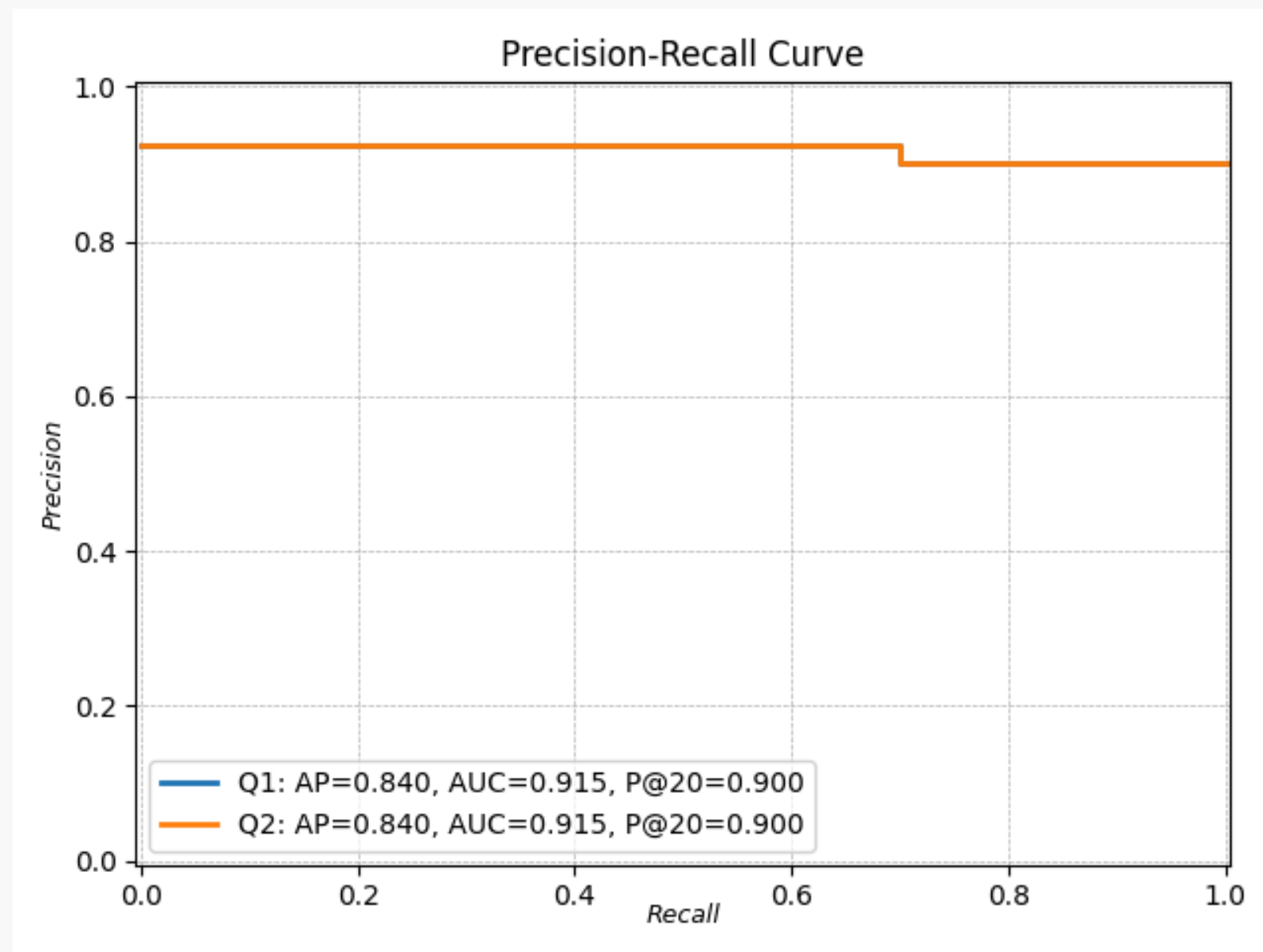
Queries and Retrieval

Information Need: *“Artificial Intelligence / Sentient Robots Themed Movies”*

Basic Schema



Custom Schema



Global Evaluation

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- The following table presents the **Mean Average Precision** across each Query on each type of schema.
- Even though we have a better result in the custom schemas and boosted queries, the differences are very minor.

<i>MAP</i>	Basic Schema	Custom Schema
Q1 (Baseline Query)	0.870	0.891
Q2 (Boosted Query)	0.874	0.900

Conclusion

We have not included each movie or series correspondent reviews as additional documents, as we wanted to focus on description. This is something that can be added as an extra.

The synonyms list could also be improved with increased quantity and more related terms. Also, adding some complexity to the information needs would better simulate a real use case of a retrieval system, and bring the value of a custom schema and boosted queries.

The future work would be to perform these improvements, introduce NLP and sentimental analysis to the parsing of description and reviews, and achieve a final and complete information retrieval system.



**Thank
you!**

